A First Course in Abstract Algebra

Section 36: Sylow Theorems

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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36 Sylow Theorems

For a finite group of order $p^a m$, where $p \nmid m$, a Sylow p-subgroup has order p^a; its number n_p divides m and satisfies $n_p \equiv 1 \pmod{p}$. A unique Sylow subgroup is normal because conjugation preserves its order.

Exercise 36.1: The Sylow 3-order when $|G| = 12$

Factor $12 = 2^2 \cdot 3$. A Sylow 3-subgroup has the largest power of 3 dividing $|G|$, namely $3^1 = 3$, as its order. This is the subgroup's order, not the number of such subgroups.

Exercise 36.2: The Sylow 3-order when $|G| = 54$

Since $54 = 2 \cdot 3^3$, the largest power of 3 dividing 54 is 27. Hence every Sylow 3-subgroup has order 27.

Exercise 36.3: Counting Sylow 2-subgroups for order 24

Sylow's counting theorem gives $n_2 \mid 24/8 = 3$ and $n_2 \equiv 1 \pmod{2}$. Both divisors 1 and 3 satisfy these conditions, so the blanks are 1 and 3. The theorem gives possible values, without asserting that both occur in a particular group.

Exercise 36.4: Counting Sylow subgroups for order 255

For $p = 3$, $n_3 \mid 85$ and $n_3 \equiv 1 \pmod{3}$. Of $1, 5, 17, 85$, only $1, 85$ qualify. For $p = 5$, $n_5 \mid 51$ and $n_5 \equiv 1 \pmod{5}$; of $1, 3, 17, 51$, only $1, 51$ qualify. Thus the requested alternatives are $n_3 = 1$ or 85, and $n_5 = 1$ or 51. Only the counting theorem is being used here, as requested.

Exercise 36.5: Sylow 3-subgroups of S_4

A Sylow 3-subgroup has order 3, since $24 = 3 \cdot 8$. Every nonidentity element in one is a 3-cycle; each 3-cycle and its inverse generate the same subgroup. The full list is

$$\langle (123) \rangle, \quad \langle (124) \rangle, \quad \langle (134) \rangle, \quad \langle (234) \rangle.$$

There are eight 3-cycles, two per subgroup, so nothing is missing. Conjugation relabels a cycle: $\sigma(ijk)\sigma^{-1} = (\sigma(i)\,\sigma(j)\,\sigma(k))$. Starting from $\langle (123) \rangle$, conjugate by (34), (234), and (1234) to get the other three subgroups respectively. Thus all four are conjugate explicitly.

Exercise 36.6: Two conjugate Sylow 2-subgroups of S_4

Take $P = \langle (1234), (13) \rangle$. With $r = (1234)$, $s = (13)$, we have $r^4 = s^2 = e$ and $srs = r^{-1}$. Its eight distinct elements are

$$e, (1234), (13)(24), (1432), (13), (24), (14)(23), (12)(34).$$

Since 8 is the largest power of 2 dividing 24, P is Sylow. Let $\tau = (34)$ and set $Q = \tau P \tau^{-1} = \langle (1243), (14) \rangle$. It again has eight elements and is Sylow. It differs from P because $(1243) \notin P$. The displayed equation itself demonstrates that P and Q are conjugate, rather than merely appealing to the general theorem.

Exercise 36.7: Definition of a p-group

An element need not have order exactly p. The correct definition is: G is a p-group if every element has order a power p^k with $k \geq 0$ (the identity has order $p^0 = 1$). For finite G this is equivalent to $|G|$ being a power of p. For example $\mathbb{Z}_{p^2}$ is a p-group and has an element of order p^2.

Exercise 36.8: Definition of the normalizer

The normalizer is a set of *elements of* G, not a set of automorphisms:

$$N_G(H) = \{g \in G : gHg^{-1} = H\}.$$

Equivalently it consists of those elements whose induced inner automorphism preserves H. Distinct group elements can induce the same inner automorphism, so replacing elements by automorphisms changes the set and potentially its order.

Exercise 36.9: Definition of a Sylow subgroup

For a finite group with $|G| = p^a m$ and $p \nmid m$, a Sylow p-subgroup is a subgroup of order p^a. Equivalently it is a maximal p-subgroup under inclusion. Say *a* Sylow subgroup, since there may be several. "The largest" suggests a unique subgroup containing all others, which usually does not exist.

Exercise 36.10: True or false: Sylow theory

(a) True: this is Sylow conjugacy.

(b) True: $n_5 \mid 3$ and $n_5 \equiv 1 \pmod 5$ force $n_5 = 1$.

(c) True by the definition of a Sylow p-subgroup.

(d) False: $\{0, 2\} < \mathbb{Z}_4$ is a 2-subgroup but is not Sylow; the entire group has order 4.

(e) True: conjugation is trivial in an abelian group, and Sylow subgroups are conjugate.

(f) False: in S_3, the normalizer of $\langle (12) \rangle$ equals $\langle (12) \rangle$, which is not normal in S_3.

(g) True: every element of the normalizer conjugates H onto itself.

(h) True: uniqueness implies conjugation invariance, and normality combined with Sylow conjugacy implies uniqueness.

(i) False: in an abelian group $N_G(H) = G$, which can strictly contain H.

(j) False: the whole group is its Sylow p-subgroup.

Exercise 36.11: The normalizer is a subgroup

Certainly $eHe^{-1} = H$. If $aHa^{-1} = H$ and $bHb^{-1} = H$, then $(ab)H(ab)^{-1} = a(bHb^{-1})a^{-1} = H$. Also $aHa^{-1} = H$ implies $a^{-1}Ha = H$, by conjugating the equality with a^{-1}. Hence the normalizer contains the identity and is closed under products and inverses, so it is a subgroup.

Exercise 36.12: A unique proper Sylow subgroup is normal

Let P be the only Sylow p-subgroup. For each $g \in G$, conjugation is an isomorphism, so gPg^{-1} has the same order as P and is again Sylow. Uniqueness gives $gPg^{-1} = P$. Thus $P \trianglelefteq G$. It is nontrivial because $p \mid |G|$, and proper because a distinct prime q divides $|G|$. This is a nontrivial proper normal subgroup, so G is not simple.

Exercise 36.13: A normal order-9 subgroup for order 45

Since $45 = 3^2 \cdot 5$, Sylow 3-subgroups have order 9. Their number divides 5 and is 1 modulo 3. Of the divisors 1 and 5, only 1 is 1 modulo 3. Thus there is a unique Sylow 3-subgroup; conjugation preserves its order and hence preserves it. This is the required normal subgroup of order 9.

Exercise 36.14: Finite p-groups and prime-power orders

If $|G| = p^a$, Lagrange's theorem makes every element order divide p^a, so each order is a power of p. Conversely suppose every element order is a power of p. If a different prime q divided $|G|$, Cauchy's theorem would produce an element of order q, contradicting the assumption. Thus no prime other than p divides $|G|$, so $|G| = p^a$. The trivial group corresponds to $a = 0$.

Exercise 36.15: A Sylow normalizer is self-normalizing

Put $M = N_G(P)$. Then $P \trianglelefteq M$, and P remains a Sylow subgroup of M because $P \subseteq M \subseteq G$ leaves no room for a larger power of p. Hence P is the unique Sylow p-subgroup of M. If $g \in N_G(M)$, conjugation by g is an automorphism of M and carries P to another Sylow p-subgroup of M. Uniqueness forces $gPg^{-1} = P$, so $g \in M$. This proves $N_G(M) \subseteq M$. Every subgroup normalizes itself, giving the reverse inclusion. Therefore $N_G(N_G(P)) = N_G(P)$.

Exercise 36.16: Conjugating a p-subgroup into a fixed Sylow subgroup

Let the p-group H act by left multiplication on the left-coset set G/P. Its size $[G : P]$ is not divisible by p. The orbit congruence therefore implies that the number of cosets fixed by all of H is nonzero modulo p, hence positive. Choose a fixed coset xP. For every $h \in H$, $hxP = xP$, so $x^{-1}hx \in P$. Thus $x^{-1}Hx \subseteq P$. Taking $g = x^{-1}$ yields the requested $gHg^{-1} \subseteq P$.

Exercise 36.17: A normal order-125 subgroup

The group order is $(35)^3 = 5^3 \cdot 7^3$. A Sylow 5-subgroup has order 125. Its number divides 343, whose divisors are $1, 7, 49, 343$; their residues modulo 5 are $1, 2, 4, 3$. Only 1 is allowed by Sylow's congruence.

Therefore the order-125 subgroup is unique and hence normal.

Exercise 36.18: Nonsimplicity at order 255

A Sylow 17-subgroup has order 17. Its number divides 15 and is 1 modulo 17. The divisors $1, 3, 5, 15$ include only one such number, namely 1. Thus the subgroup is unique, normal, nontrivial, and proper. Consequently no group of order 255 is simple.

Exercise 36.19: The small-cofactor criterion and its prime exception

Qualification of the printed statement. As written it includes $m = 1, r = 1$, so it is false: $\mathbb{Z}_p$ is simple and has order $p^1 \cdot 1$ with $1 < p$. For the intended composite-order assertion, let $|G| = p^r m$ with $1 < m < p$. Since $p \nmid m$, a Sylow p-subgroup has order p^r. Its number divides m and is congruent to 1 modulo p. But every positive divisor of m lies between 1 and $m < p$, so only 1 is possible. The unique Sylow subgroup is nontrivial, normal, and proper. If $m = 1$ and $r > 1$, the group is a p-group. Its center is nontrivial by Exercise 36.20. If the center is proper it is the required normal subgroup; if the group is abelian, any order-p subgroup is normal and proper. Thus only the prime-order case must be excluded.

Exercise 36.20: Conjugation fixed points and the center

(a) A point $x \in G$ is fixed by every conjugation precisely when $gxg^{-1} = x$ for every g, equivalently $gx = xg$ for every g. Thus the fixed-point set is $Z(G)$. **(b)** When $|G| = p^n$ with $n \geq 1$, the orbit congruence for this action gives $|Z(G)| \equiv |G| \equiv 0 \pmod{p}$. The center contains e, so its order is a positive multiple of p and in particular at least p. Hence it is nontrivial.

Exercise 36.21: Normal subgroups of every prime-power order

We prove the stronger assertion of a nested normal chain by induction on n. For $n = 0$, take $H_0 = G = 1$. For $n \geq 1$, the center is nontrivial, so Cauchy's theorem gives a central subgroup C of order p. Centrality makes $C \trianglelefteq G$. The quotient G/C has order p^{n-1}. By induction it has a nested chain of subgroups $\overline{H}_j \trianglelefteq G/C$ of orders p^j for $0 \leq j \leq n-1$. If $\pi : G \to G/C$, set $H_0 = 1$ and $H_{j+1} = \pi^{-1}(\overline{H}_j)$. Preimages of normal subgroups are normal, and each preimage has order $|C|\,|\overline{H}_j| = p^{j+1}$. They are nested because the quotient terms are nested. This gives all required H_i, with $H_n = G$.

Exercise 36.22: Normal p-subgroups lie in every Sylow subgroup

Let S be any Sylow p-subgroup. Since $P \trianglelefteq G$, PS is a subgroup. Its order is

$$|PS| = \frac{|P||S|}{|P \cap S|},$$

a power of p, since all three orders are powers of p. It contains S, which already has the largest possible p-power order in G. Thus $PS = S$, and hence $P \subseteq S$. Since S was arbitrary, this holds for every Sylow p-subgroup.